Abhishek Kumar Singh

Embedded Systems | Edge AI | IoT
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PROFILE

ECE student specialising in IoT, with a C++ early-exit inference runtime achieving approximately 25% lower mean latency in a controlled CPU benchmark. Hands-on ESP32 sensing, signed telemetry and controller testing; seeking embedded systems and Edge AI internships.

EDUCATION

Madan Mohan Malaviya University of Technology, Gorakhpur

B.Tech, Electronics & Communication Engineering (IoT) | 2025-2029
First-year CGPA: **8.4/10**

KV IIT Kanpur | Class XII: **89%** | Class X: **89.4%**

SELECTED PROJECTS

AeroNode | Adaptive Edge-AI Inference Runtime

github.com/abhishekkr10/AeroNode

- Engineered staged C++17 inference with persistent ONNX Runtime sessions and confidence-based early exits, skipping unnecessary stages while checking predictions against reference execution.
- Measured **25.2% lower mean latency**: 2.1052 ms versus 2.8139 ms; accuracy 72.6857% versus 72.9400% on 1,966 held-out images. Windows CPU, batch 1, one thread, three repetitions.

SafeTrack (TrackSure) | Signed IoT Shipment Monitoring

github.com/inamansingh2401/tamperproject

- Integrated ESP32-S3 firmware with DHT11, MPU6050 and light sensing; used ATECC508A signatures and a linked event history to support tamper-evident shipment records.
- Connected device events through a Python gateway with a durable **SQLite outbox** to a FastAPI service, **PostgreSQL storage** and Admin/User dashboards. Hardware prototype.

CausalBench | Reproducible Controller-Failure Lab

github.com/abhishekkr10/CausalBench

- Created a local lab to reproduce sensor/controller failures using timestamped traces, seeded faults and baseline-versus-guarded decision replay with simulated actuation.
- Automated failure minimization, regression checks and source/evidence exports; repository documents local validation with **93 Python tests and 13 browser workflows**.

JOCKY | Defensive Investigation Tooling

github.com/abhishekkr10/jocky-forensics

Rust investigation compiler and evidence analysis with signed integrity checks; working local platform with a synthetic public demonstration.

TECHNICAL SKILLS

Embedded/Hardware: C, C++17, ESP32-S3, I2C, UART, MPU6050, DHT11, ATECC508A

ML/Inference: Python, PyTorch, ONNX, ONNX Runtime, ZeroMQ

Backend/Tools: FastAPI, PostgreSQL, SQLite, React, TypeScript/JavaScript, Rust, CMake, Git/GitHub